

# Pablo Zúñiga Carvajal

Industrial & Systems Engineer, BSc & MSc Equivalent

Data Modeling · Applied Statistics · Complex Systems

Valparaíso, Chile · International Relocation Ready · ESTA & H-1B1 Visa Eligible

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## Technical Skills

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- **Code Languages & Advanced Analytics:** Python (FastAPI, PySpark, Polars), R (Tidyverse, Survival, Forecast), SQL, Statistical & Inferential Modeling, Time-Series Analysis, Linear & Integer Programming, Stochastic Simulation & Monte Carlo Methods
- **Enterprise Data Architecture & Ontologies:** ETL/ELT Pipelines, Metadata & Ontology Modeling, Data Governance, Enterprise Lineage, Relational Databases, Spatial Data Infrastructure (.gpx, .fit, .kmz), API REST
- **Infrastructure, Cloud & Orchestration:** Git/GitHub, Cloud Systems (MS Azure), Environment Orchestration, Distributed Protocols (Web3/On-Chain), Object-Oriented Programming (OOP)
- **Systems Architecture & Technical Tooling:** Complex Systems Modeling, Process Engineering (BPMN), UML Diagraming, Knowledge Architectures, Technical Documentation (Quarto, Markdown, LaTeX, HTML/CSS)

## Work Experience

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### Cuxhaven Labs | Principal Data & Systems Consultant | Remote

May 2026 – Present

Architected production-grade ETL/ELT pipelines and metadata ontologies for critical infrastructure under strict auditability standards. Engineered operational resilience frameworks and disaster recovery protocols to guarantee continuous operational uptime during system incidents.

### Independent Forward-Deployed Engineering Consultant | Remote

Jul 2022 – Present

**Telemetry Analytics Pipeline:** Built an end-to-end time-series analytics pipeline in Python/R processing pace-distance telemetry across 1.8k+ athletes to quantify physiological gaps and optimize pacing strategies under multi-variable environmental conditions. Designed relational database structures, automated data flows, and technical documentation architectures using Git/Markdown.

### Decentralized Systems & Protocols | On-Chain Operations Specialist | Remote

Jan 2020 – Jul 2022

Managed multi-sig treasury architecture (\$50k+) across Polkadot, Ethereum, and BSC. Executed dynamic gas optimization, cross-chain messaging, and smart contract coordination for 1,100+ concurrent users.

### Banco de Crédito e Inversiones | Process Engineer (Data Governance) | Santiago, Chile

May 2019 – Sep 2019

Orchestrated automated metadata integration pipelines from core transactional systems into MS Azure, standardizing BPMN workflows. Audited data architecture compliance in regulated banking environments, enforcing schema consistency and RBAC security controls.

### ESVAL (Water Utility) | Lead Data & Project Engineer | Valparaíso, Chile

Jul 2017 – Dec 2017

Developed a spatio-temporal risk model using Cox Proportional Hazards in R across 5,300 km of pipeline serving 1.9M+ residents. Delivered predictive maintenance risk scores via Agile/PMO frameworks to prioritize capital expenditure allocation for regional grid resilience.

### ENAMI & Antofagasta Minerals | Industrial Engineering Intern | Chile

2014 & 2016

Audited metallurgical inventory database (75k+ SKUs), uncovering 2.5M USD in accounting discrepancies for corporate financial audits. Monitored open-pit mining geotechnical telemetry for slope failure prediction and early risk mitigation in mission-critical environments.

## Education

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### Pontificia Universidad Católica de Valparaíso | Valparaíso, Chile

Industrial & Systems Engineering (Integrated 6-Year Program – BSc & MSc Equivalent) | 2011 – 2017

Core Academic Foundations: Statistics, Systems Theory, Operations Research, Econometrics, Systems Modeling.

## Leadership & Teaching

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### Graduate Teaching Assistant (PUCV)

Instructed Applied Data Science, MIS, and Quantitative Mathematics for 100+ engineering students (2018–2021).

### Technical Instructor (BancoEstado / UTFSM)

Delivered time-series analytics and high-frequency financial modeling training to institutional trading desk staff.